

Contestant Number: _____

Time: _____

Rank: _____



COMPUTER SECURITY

(320)

STATE – 2020

TOTAL POINTS

_____ (500 Points)

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

TEST TIME: 60 Minutes

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Identify the letter of the choice that best completes the statement or answers the question. Mark A if the statement is true. Mark B if the statement is false.

- 1) Which of the following terms indicates that information is to be read, only by those people for whom it is intended?
 - a) confidentiality
 - b) integrity
 - c) availability
 - d) accounting
- 2) What authentication type is the default for Active Directory?
 - a) NTLM
 - b) Kerberos
 - c) MS-CHAP
 - d) MS-CHAPv2
- 3) What do you use to define how long a password is in Windows?
 - a) registry
 - b) Users applet in the Control Panel
 - c) group policies
 - d) NTFS files
- 4) Which type of system detects unauthorized intruders and then takes action to stop them from proceeding?
 - a) IDS
 - b) IPS
 - c) VLAN
 - d) NAT
- 5) What Windows feature notifies you when something tries to make changes to your computer without your knowledge?
 - a) WDS
 - b) NAT
 - c) Windows Defender
 - d) UAC
- 6) If using the MD5 hashing algorithm, what is the length to which each message is padded?
 - a) 32 bits
 - b) 64 bits
 - c) 128 bits
 - d) 512 bits
- 7) What type of additional attack does ARP spoofing rely on?
 - a) DNS Poisoning
 - b) replay
 - c) MITB
 - d) MAC spoofing

- 8) An administrator has two servers that host the same web content, but only one server is utilized at a given time. What can be configured that can help to evenly distribute work across the network, and make use of both servers in a manner that is transparent to the end users?
 - a) stateful packet filtering
 - b) Load balancing
 - c) failover servers
 - d) proxy servers
- 9) A firewall that keeps a record of the state of a connection between an internal computer and an external device is using what technology below?
 - a) Stateful frame filtering
 - b) Stateless frame filtering
 - c) Stateful packet filtering
 - d) Stateless packet filtering
- 10) When using SNMPv1 or SNMPv2, what piece of information is needed to view information from an agent?
 - a) entity
 - b) community string
 - c) MIB
 - d) OID
- 11) DNS poisoning can be prevented using the latest edition of what software below?
 - a) BIND
 - b) DHCP
 - c) WINS
 - d) finger
- 12) What is the first step in computer forensics chain of custody?
 - a) Reporting
 - b) Examination
 - c) Analyze
 - d) Collection
- 13) One of the armored virus infection techniques utilizes encryption to make virus code more difficult to detect, in addition to separating virus code into different pieces and inject these pieces throughout the infected program code. What is the name for this technique?
 - a) stealth
 - b) appender
 - c) Swiss cheese
 - d) split

- 14) What is the name of the process that basically takes a snapshot of the current security of an organization?
- a) threat analysis
 - b) vulnerability appraisal
 - c) risk assessment
 - d) threat assessment
- 15) What is the end result of a penetration test?
- a) penetration test profile
 - b) penetration test report
 - c) penetration test system
 - d) penetration test view
- 16) Select the vulnerability scan type that will use only the available information to hypothesize the status of the vulnerability.
- a) active
 - b) non-intrusive
 - c) passive
 - d) intrusive
- 17) What access control model below is considered to be the most restrictive access control model, and involves assigning access controls to users strictly according to the custodian?
- a) Mandatory Access Control
 - b) Role Based Access Control
 - c) Discretionary Access Control
 - d) Rule Based Access Control
- 18) Which type of biometrics is based on the perception, thought process, and understanding of the user?
- a) cognitive biometrics
 - b) reactive biometrics
 - c) standard biometrics
 - d) physical biometrics
- 19) What type of management system below can help distribute and manage public and corporate apps?
- a) Wireless Device Management
 - b) Mobile Device Management
 - c) Total Device Management
 - d) Extended Device Management
- 20) Which of the following is an example of biometric security?
- a) Standard scan
 - b) Cognitive scan
 - c) Reactive recognition
 - d) Facial recognition

- 21) What security concept states a user should only be given the minimum set of permissions required to perform necessary tasks?
- a) least functionality
 - b) whitelisting
 - c) blacklisting
 - d) function limiting
- 22) Select the option that best describes a policy:
- a) A collection of requirements specific to the system or procedure that must be met by everyone
 - b) A collection of suggestions that should be implemented
 - c) A list of all items that have a positive economic value
 - d) A document that outlines specific requirements or rules that must be met
- 23) Generally considered to be the most important information security policies, what item below defines the actions a user may perform while accessing systems and networking equipment?
- a) acceptable use policies
 - b) encryption policies
 - c) data loss policies
 - d) VPN policies
- 24) What do spammers and hackers look for when they want to send email through your network?
- a) open SMTP servers
 - b) open web servers
 - c) open POP3 servers
 - d) open FTP servers
- 25) Which settings options in Google Chrome would you use to delete history and cookies?
- a) Advanced
 - b) Privacy
 - c) Security
 - d) Settings
- 26) What technique is used to send you to a fake, but realistic-looking, website to verify your account information?
- a) spoofing
 - b) smurfing
 - c) man-in-the-middle
 - d) phishing
- 27) What type of server would you install that would be used to trap a hacker?
- a) honeypot
 - b) NAT
 - c) IPS
 - d) IDS

- 28) How many firewalls would you use to create a sandwich DMZ?
- a) 1
 - b) 2
 - c) 3
 - d) 4
- 29) What special area serves as a buffer area between the Internet and the internal network and can be used to hold web servers that are accessed from the Internet?
- a) DMZ
 - b) NAT
 - c) VLAN
 - d) PLC
- 30) Which of the following is a form of ransomware?
- a) Crypto-malware
 - b) Systemic Ransom
 - c) Port Holder
 - d) Port blocker
- 31) You have several Internet web servers that need to communicate with a SQL server. Where would you place the SQL server?
- a) internal network
 - b) DMZ
 - c) Internet
 - d) isolated VLAN
- 32) What technology allows a user at home to connect to the corporate network?
- a) NAT
 - b) VPN
 - c) DMZ
 - d) PLC
- 33) What settings are used to keep track of incorrect logon attempts and lock the account if too many attempts are detected within a certain set time?
- a) password policy
 - b) authentication tracker
 - c) account lockout
 - d) user parameters
- 34) What type of software can you use to view usernames and passwords broadcasted over the network?
- a) dictionary software
 - b) password leaker
 - c) keylogger
 - d) sniffer

- 35) Where are users and permissions stored for an NTFS folder?
- a) access log
 - b) access file
 - c) registry
 - d) ACL
- 36) Which authentication sends the username and password in plain text?
- a) MS-CHAP
 - b) CHAP
 - c) PAP
 - d) SPAP
- 37) In Windows, what do you use to enable auditing?
- a) registry
 - b) group policies
 - c) NTFS permissions
 - d) access log
- 38) Which of the following makes sure that data is not changed when it is not supposed to be?
- a) confidentiality
 - b) integrity
 - c) availability
 - d) accounting
- 39) Which of the following is **not** a response when dealing with a risk?
- a) avoidance
 - b) mitigation
 - c) transfer
 - d) patching
- 40) The CIA triad consists of Confidentiality, _____, and Availability.
- a) Intensity
 - b) Integrity
 - c) Integral
 - d) Independent
- 41) In dealing with risks, which response is done by buying insurance to protect your bottom line if such a disaster or threat is realized?
- a) risk avoidance
 - b) risk acceptance
 - c) risk mitigation
 - d) risk transfer
- 42) What kind of policy defines the actions users may perform while accessing systems and networking equipment?
- a) VPN access policy
 - b) network use policy
 - c) privacy use policy
 - d) acceptable use policy

- 43) In Microsoft Windows, what type of templates are a collection of security configuration settings?
- a) security
 - b) user
 - c) system
 - d) administrator
- 44) What type of filtering software divides email messages that have been received into two piles, spam and non-spam, and then analyzes every word in each email and determines how frequently a word occurs in the spam pile compared to the non-spam pile?
- a) Blacklist filtering
 - b) Whitelist filtering
 - c) Bayesian filtering
 - d) Extension filtering
- 45) How could an administrator initially manage applications on mobile devices using a technique called “app wrapping”?
- a) Mobile Application Management
 - b) Extended Application Management
 - c) Remote Application Management
 - d) Cloud Application Management
- 46) The use of a single authentication credential that is shared across multiple networks is called:
- a) access management
 - b) authorization management
 - c) identity management
 - d) risk management
- 47) What type of penetration testing technique is used if the tester has no prior knowledge of the network infrastructure that is being tested?
- a) white box
 - b) gray box
 - c) black box
 - d) sealed box
- 48) What type of computer code is typically added to a legitimate program but lies dormant until a specific logical event triggers it?
- a) script
 - b) logic bomb
 - c) macro virus
 - d) metamorphic virus

- 49) What type of undocumented, yet benign, hidden feature launches after a special set of commands, key combinations, or mouse clicks?
- a) Trojan horse
 - b) virus
 - c) bug
 - d) Easter egg
- 50) What is the port number for HTTPS?
- a) 80
 - b) 8080
 - c) 23
 - d) 443